Study Notes: Chemistry - G9

UNIT 2: MEASUREMENTS AND SCIENTIFIC METHODS: 2.1 Measurements and Units in Chemistry, 2.2 Chemistry as Experimental Science, UNIT 2: Unit Summary

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2.1 Measurements and Units in Chemistry

This chapter provides a comprehensive overview of measurements in chemistry, covering the International System of Units (SI), derived units, prefixes, uncertainty in measurements, precision and accuracy, significant figures, scientific notation, and the scientific method as applied in experimental chemistry. It also delves into practical laboratory skills, safety protocols, and proper laboratory report writing.

1. Introduction to Measurement

- **Definition of Measurement:** Measurement is the comparison of a physical quantity to be measured with a unit of measurement, which is a fixed standard.
  - *Example:* On a centimeter scale, the centimeter unit is the standard of comparison.
- **Traditional Methods:** In traditional markets, people often estimate sizes or use indigenous measurement methods for goods.
  - *Figure 2.1:* Illustrates traditional markets where goods are exchanged using estimated sizes or indigenous methods.
- **Importance in Chemistry:** The study of chemistry relies heavily on accurate measurements.
  - Chemists use measurements to:
    - Compare properties of different substances.
    - Assess changes resulting from experiments.
- **Common Measuring Devices:**
  - **Length:** Meter stick
  - **Volume:** Burette, pipette, graduated cylinder, volumetric flask
  - **Mass:** Balance
  - **Temperature:** Thermometer
  - *Figure 2.2:* Displays various common measuring devices found in a chemistry laboratory.
- **Macroscopic vs. Microscopic Properties:**
  - **Macroscopic properties:** Can be determined directly (e.g., using the instruments listed above).
  - **Microscopic properties:** (on the atomic or molecular scale) must be determined by indirect methods.
- **Importance of Units:** A measured quantity is meaningless without an appropriate unit.
  - *Example:* Stating a distance is “275” is unclear; specifying “275 kilometers” makes it meaningful. In science, units are essential for correct measurement statements.

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📝 Practice Question 1 / የልምምድ ጥያቄ 1 (EUEE-style)

Question / ጥያቄ: What is *measurement*? / *መለካት ምንድን ነው*?

Options / አማራጮች:

- A. Estimating the sizes of goods only by traditional methods [የሸቀጦችን መጠን በባህላዊ ዘዴዎች ብቻ መገመት]
- B. The study of the structure of atoms [የአቶሞችን አወቃቀር ማጥናት]
- C. The comparison of a physical quantity to be measured with a unit of measurement, which is a fixed standard [ሊለካ የሚፈለግን አካላዊ ብዛት ከቋሚ መመዘኛ ከሆነው የመለኪያ አሃድ ጋር ማወዳደር]
- D. The calculation of density only [ጥግግትን ብቻ ማስላት]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Measurement is the comparison of a physical quantity with a unit, which is a fixed standard; for example, the centimeter is the standard on a centimeter scale. Option A is the main distractor: estimating sizes is a traditional method used in markets, not the definition of measurement.

ማብራሪያ (አማርኛ): መለካት አካላዊ ብዛትን ከቋሚ መመዘኛ ከሆነ የመለኪያ አሃድ ጋር ማወዳደር ነው፤ ለምሳሌ በሴንቲሜትር መለኪያ ላይ ሴንቲሜትር መመዘኛ ነው። አማራጭ A አሳሳች ነው፤ መጠንን መገመት በገበያዎች የሚጠቀሙበት ባህላዊ ዘዴ እንጂ የመለካት ትርጉም አይደለም።

📝 Practice Question 2 / የልምምድ ጥያቄ 2 (EUEE-style)

Question / ጥያቄ: Which measuring device is correctly matched with the quantity it measures? / የትኛው የመለኪያ መሣሪያ ከሚለካው ብዛት ጋር በትክክል ተዛምዷል?

Options / አማራጮች:

- A. Balance – volume [ሚዛን – መጠን (ቮልዩም)]
- B. Thermometer – temperature [ቴርሞሜትር – የሙቀት መጠን]
- C. Meter stick – mass [ሜትር ዱላ – ክብደት (ማስ)]
- D. Burette – length [ቡሬት – ርዝመት]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): The guide pairs the thermometer with temperature, the balance with mass, the meter stick with length, and the burette, pipette, graduated cylinder and volumetric flask with volume. Option A is the main distractor, because a balance measures mass, not volume.

ማብራሪያ (አማርኛ): መመሪያው ቴርሞሜትርን ከሙቀት መጠን፣ ሚዛንን ከክብደት፣ የሜትር ዱላን ከርዝመት፣ ቡሬት፣ ፒፔት፣ የተመረቀ ሲሊንደርና የቮልዩሜትሪክ ብልቃጥን ደግሞ ከመጠን (ቮልዩም) ጋር ያዛምዳል። አማራጭ A አሳሳች ነው፤ ሚዛን የሚለካው ክብደትን እንጂ መጠንን አይደለም።

📝 Practice Question 3 / የልምምድ ጥያቄ 3 (EUEE-style)

Question / ጥያቄ: Which statement about macroscopic and microscopic properties is correct? / ስለ ማክሮስኮፒክና ማይክሮስኮፒክ ባህሪያት የትኛው መግለጫ ትክክል ነው?

Options / አማራጮች:

- A. Microscopic properties can be determined directly with a meter stick [ማይክሮስኮፒክ ባህሪያት በሜትር ዱላ በቀጥታ ሊወሰኑ ይችላሉ]
- B. Macroscopic properties must always be determined by indirect methods [ማክሮስኮፒክ ባህሪያት ሁልጊዜ በተዘዋዋሪ ዘዴዎች መወሰን አለባቸው]
- C. Both macroscopic and microscopic properties must be determined by indirect methods [ማክሮስኮፒክም ማይክሮስኮፒክም ባህሪያት በተዘዋዋሪ ዘዴዎች መወሰን አለባቸው]
- D. Macroscopic properties can be determined directly with instruments, while microscopic (atomic or molecular scale) properties must be determined by indirect methods [ማክሮስኮፒክ ባህሪያት በመሣሪያዎች በቀጥታ ሊወሰኑ ይችላሉ፤ ማይክሮስኮፒክ (በአቶም ወይም ሞለኪውል ደረጃ) ባህሪያት ግን በተዘዋዋሪ ዘዴዎች መወሰን አለባቸው]

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Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Macroscopic properties can be determined directly with measuring instruments, but microscopic properties on the atomic or molecular scale must be determined by indirect methods. Option C is the main distractor, because it wrongly treats macroscopic properties as indirect too.

ማብራሪያ (አማርኛ): ማክሮስኮፒክ ባህሪያት በመለኪያ መሣሪያዎች በቀጥታ ሊወሰኑ ይችላሉ፤ በአቶም ወይም ሞለኪውል ደረጃ ያሉ ማይክሮስኮፒክ ባህሪያት ግን በተዘዋዋሪ ዘዴዎች መወሰን አለባቸው። አማራጭ C ማክሮስኮፒክ ባህሪያትንም በተዘዋዋሪ ዘዴ እንደሚወሰኑ በስህተት ስለሚያሳይ አሳሳች ነው።

2. SI Units (The International System of Units)

- **Background:**
  - For many years, scientists used metric units (related decimally by powers of 10).
  - In 1960, the General Conference of Weights and Measures (the international authority on units) proposed a revised metric system called the **International System of Units (SI Units)**.
- **SI Base Units:** The SI system is built upon seven fundamental base units.
  - Measurements frequently used in chemistry include time, mass, volume, density, and temperature.

Base Quantity	Name of Unit	Symbol
Length	Meter	m
Mass	Kilogram	kg
Time	Second	s
Electrical current	Ampere	A
Temperature	Kelvin	K
Amount of substance	Mole	mol
Luminous intensity	Candela	cd

📝 Practice Question 4 / የልምምድ ጥያቄ 4 (EUEE-style)

Question / ጥያቄ: What is the SI base unit of mass? / የክብደት (ማስ) የኤስአይ (SI) መሠረታዊ አሃድ ምንድን ነው?

Options / አማራጮች:

- A. Kilogram (kg) [ኪሎግራም (kg)]
- B. Gram (g) [ግራም (g)]
- C. Milligram (mg) [ሚሊግራም (mg)]
- D. Pound (lb) [ፓውንድ (lb)]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): In the table of SI base units, the unit of mass is the kilogram (kg). Gram is the main distractor: it is related to the kilogram (1 kilogram = 1,000 grams) but is not the SI base unit.

ማብራሪያ (አማርኛ): በኤስአይ መሠረታዊ አሃዶች ሰንጠረዥ ውስጥ የክብደት አሃድ ኪሎግራም (kg) ነው። ግራም ከኪሎግራም ጋር የሚዛመድ (1 ኪሎግራም = 1,000 ግራም) ቢሆንም መሠረታዊ የኤስአይ አሃድ ስላልሆነ አሳሳች አማራጭ ነው።

📝 Practice Question 5 / የልምምድ ጥያቄ 5 (EUEE-style)

Question / ጥያቄ: Which of the following is **NOT** one of the seven SI base units? / ከሚከተሉት ውስጥ ከሰባቱ የኤስአይ መሠረታዊ አሃዶች አንዱ ያልሆነው የትኛው ነው?

Options / አማራጮች:

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- A. Meter [ሜትር]
- B. Second [ሰከንድ]
- C. Mole [ሞል]
- D. Liter [ሊትር]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): The seven SI base units are the meter, kilogram, second, ampere, kelvin, mole and candela. The liter is a traditional unit equal to a cubic decimeter, so it is not a base unit; mole is the main distractor, since it is the base unit for the amount of substance.

ማብራሪያ (አማርኛ): ሰባቱ የኤስአይ መሠረታዊ አሃዶች ሜትር፣ ኪሎግራም፣ ሰከንድ፣ አምፒር፣ ኬልቪን፣ ሞልና ካንደላ ናቸው። ሊትር ከኪዩቢክ ዴሲሜትር ጋር እኩል የሆነ ባህላዊ አሃድ ስለሆነ መሠረታዊ አሃድ አይደለም፤ ሞል የንጥረ ነገር መጠን መሠረታዊ አሃድ ስለሆነ አሳሳች አማራጭ ነው።

2.1 Heat and Temperature

- **Temperature:** Measures the *intensity of heat*, or the "hotness" or "coldness" of a body.
- **Heat:** A form of energy that always flows spontaneously from a hotter body to a colder body.
- **Temperature Scales:**
  - **Celsius (°C) and Kelvin (K):**
    - Between the freezing point and boiling point of water, there are 100 steps on both Celsius and Kelvin scales.
    - The "degree" size is the same on both scales.
    - Every Kelvin temperature is 273.15 units above the corresponding Celsius temperature.
    - **Relationship:**
      - K = °C + 273.15
      - °C = K − 273.15
    - In SI, "degrees Kelvin" are abbreviated as K and called **kelvins**.
    - A temperature change has the same numerical value on Celsius and Kelvin scales (e.g., a 34 °C change is also a 34 K change).
  - **Fahrenheit (°F) and Celsius (°C):**
    - Intervals between the same reference points are 180 °F and 100 °C, respectively.
    - A Fahrenheit degree is smaller than a Celsius degree (180 °F covers the same interval as 100 °C).
    - **Unit Factors for Temperature Changes:**
      - 180 °F / 100 °C or 1.8 °F / 1.0 °C
      - 100 °C / 180 °F or 1.0 °C / 1.8 °F
    - **Conversion Formulas:** The starting points (zero points) of the scales are different.
      - °C = (1.0 °C / 1.8 °F) × (°F − 32 °F)
      - °F = (1.8 °F / 1.0 °C) × °C + 32 °F
  - *Figure 2.3:* Illustrates the relationships among the Kelvin, Celsius, and Fahrenheit temperature scales.

Example 2.1: Temperature Conversion

- **Problem:** When the temperature reaches "100. °F in the shade," what is this temperature on the Celsius scale?
- **Solution:**
  - Use the relationship: °C = (1.0 °C / 1.8 °F) × (°F − 32 °F)
  - Substitute 100 °F:
    - °C = (1.0 / 1.8) × (100 − 32) = (1.0 / 1.8) × (68) ≈ 38 °C
  - **Answer:** 38 °C

📝 Practice Question 6 / የልምምድ ጥያቄ 6 (EUEE-style)

Question / ጥያቄ: In which direction does heat always flow spontaneously? / ሙቀት ሁልጊዜ በራሱ የሚፈሰው በየትኛው አቅጣጫ ነው?

Options / አማራጮች:

- A. From a colder body to a hotter body [ከቀዝቃዛ አካል ወደ ሞቃት አካል]
- B. From a hotter body to a colder body [ከሞቃት አካል ወደ ቀዝቃዛ አካል]
- C. Equally in both directions [በሁለቱም አቅጣጫ እኩል]
- D. Only from solids to liquids [ከጠጣር ወደ ፈሳሽ ብቻ]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): Heat is a form of energy that always flows spontaneously from a hotter body to a colder body. Option A is the main distractor, since it reverses the direction of flow.

ማብራሪያ (አማርኛ): ሙቀት ሁልጊዜ በራሱ ከሞቃት አካል ወደ ቀዝቃዛ አካል የሚፈስ የኃይል ዓይነት ነው። አማራጭ A የፍሰቱን አቅጣጫ ስለሚገለብጥ አሳሳች ነው።

📝 Practice Question 7 / የልምምድ ጥያቄ 7 (EUEE-style)

Question / ጥያቄ: What is 25 °C on the Kelvin scale? / 25 °C በኬልቪን ሚዛን ስንት ነው?

Options / አማራጮች:

- A. 25 K [25 K]
- B. 248.15 K [248.15 K]
- C. 298.15 K [298.15 K]
- D. 273.15 K [273.15 K]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Using K = °C + 273.15, the temperature is 25 + 273.15 = 298.15 K. Option B is the main distractor, because it comes from subtracting 273.15 instead of adding it.

ማብራሪያ (አማርኛ): K = °C + 273.15 በመጠቀም ሙቀቱ 25 + 273.15 = 298.15 K ነው። አማራጭ B 273.15ን ከመደመር ይልቅ በመቀነስ የሚገኝ ስለሆነ አሳሳች ነው።

📝 Practice Question 8 / የልምምድ ጥያቄ 8 (EUEE-style)

Question / ጥያቄ: The temperature of a liquid rises by 15 °C. What is this temperature change on the Kelvin scale? / የአንድ ፈሳሽ ሙቀት በ15 °C ጨመረ። ይህ የሙቀት ለውጥ በኬልቪን ሚዛን ስንት ነው?

Options / አማራጮች:

- A. 15 K [15 K]
- B. 288.15 K [288.15 K]
- C. 258.15 K [258.15 K]
- D. 27 K [27 K]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): A temperature change has the same numerical value on the Celsius and Kelvin scales, so a 15 °C change is a 15 K change. Option B is the main distractor, because it wrongly adds 273.15, which applies to temperature readings, not to changes.

ማብራሪያ (አማርኛ): የሙቀት ለውጥ በሴልሺየስና በኬልቪን ሚዛኖች ላይ ተመሳሳይ የቁጥር ዋጋ ስላለው የ15 °C ለውጥ የ15 K ለውጥ ነው። አማራጭ B 273.15ን በስህተት ይጨምራል፤ ይህ ለሙቀት ንባቦች እንጂ ለለውጦች አይሠራም፤ ስለዚህ አሳሳች ነው።

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📝 Practice Question 9 / የልምምድ ጥያቄ 9 (EUEE-style)

Question / ጥያቄ: Using °C = (1.0 °C / 1.8 °F) × (°F − 32 °F), what is 68 °F on the Celsius scale? / °C = (1.0 °C / 1.8 °F) × (°F − 32 °F) በመጠቀም 68 °F በሴልሺየስ ሚዛን ስንት ነው?

Options / አማራጮች:

- A. 154.4 °C [154.4 °C]
- B. 36 °C [36 °C]
- C. 20 °C [20 °C]
- D. 5.8 °C [5.8 °C]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Substituting gives (1.0 / 1.8) × (68 − 32) = 36 / 1.8 = 20 °C. Option B is the main distractor: it stops at 68 − 32 = 36 and forgets to apply the unit factor 1.0 °C / 1.8 °F, which is needed because a Fahrenheit degree is smaller than a Celsius degree.

ማብራሪያ (አማርኛ): በመተካት (1.0 / 1.8) × (68 − 32) = 36 / 1.8 = 20 °C ይገኛል። አማራጭ B በ68 − 32 = 36 ላይ ቆሞ የ1.0 °C / 1.8 °F ምክንያቱን መጠቀም ስለሚረሳ አሳሳች ነው፤ ይህ ምክንያት የሚያስፈልገው የፋራንሃይት ዲግሪ ከሴልሺየስ ዲግሪ ያነሰ በመሆኑ ነው።

3. Derived Units

Derived units are all other SI units of measurement that are formed by combining SI base units.

- **Formation:** Derived units are created by using base units in equations that define other physical quantities.
  - **Area:** Length multiplied by width.
    - SI unit of area = (SI unit of length) × (SI unit of width)
    - SI unit of area = meter × meter = m^2
  - **Speed:** Distance divided by time.
    - SI unit of speed = (SI unit of distance) / (SI unit of time)
    - SI unit of speed = meters per second (m/s or $m\ s^{-1}$)

Quantity	Definition of Quantity	SI Unit
Area	Length squared	m^2
Volume	Length cubed	m^3
Density	Mass per unit volume	kg/m^3
Speed	Distance per unit time	m/s
Acceleration	Speed change per unit time	m/s^2
Force	Mass times acceleration	$kg{\cdot}m/s^2$ (= newton, N)
Pressure	Force per unit area	$kg/(m{\cdot}s^2)$ (= pascal, Pa)
Energy	Force times distance	$kg{\cdot}m^2/s^2$ (= joule, J)

📝 Practice Question 10 / የልምምድ ጥያቄ 10 (EUEE-style)

Question / ጥያቄ: Which quantity is correctly matched with its SI derived unit? / የትኛው ብዛት ከኤስአይ ተወላጅ (derived) አሃዱ ጋር በትክክል ተዛምዷል?

Options / አማራጮች:

- A. Area – m^3 [ስፋት – m^3]
- B. Speed – m/s^2 [ፍጥነት – m/s^2]
- C. Pressure – pascal (Pa) [ግፊት – ፓስካል (Pa)]
- D. Energy – newton (N) [ኃይል – ኒውተን (N)]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Pressure is force per unit area and its SI unit is the pascal (Pa). The other pairs are wrong: area is m^2, speed is m/s (m/s^2 is acceleration), and energy is measured in joules (J), while the newton (N) is the unit of force.

ማብራሪያ (አማርኛ): ግፊት በአንድ አሃድ ስፋት ላይ የሚሠራ ኃይል ሲሆን የኤስአይ አሃዱ ፓስካል (Pa) ነው። ሌሎቹ ጥንዶች ስህተት ናቸው፤ ስፋት m^2 ነው፣ ፍጥነት m/s ነው (m/s^2 የፍጥነት ጭማሪ ነው)፣ ኃይል (ኢነርጂ) በጁል (J) ይለካል፤ ኒውተን (N) የ"ፎርስ" አሃድ ነው።

3.1 Volume

- **Definition:** Volume is defined as length cubed.
- **SI Unit:** Cubic meter (m^3).
  - This unit is often too large for typical laboratory work.
- **Common Laboratory Units:**
  - Cubic decimeters (dm^3)
  - Cubic centimeters (cm^3, also written as cc)
  - **Liter (L):** A traditional unit equal to a cubic decimeter.
  - **Milliliter (mL):** Commonly used for laboratory glassware (1000 mL = 1 L).
- **Relationships:**
  - 1 dm = 10 cm
  - 1 dm^3 = (10 cm)^3 = 1000 cm^3
  - Therefore:
    - 1 L = 1 dm^3
    - 1 mL = 1 cm^3

📝 Practice Question 11 / የልምምድ ጥያቄ 11 (EUEE-style)

Question / ጥያቄ: Which of the following equalities is correct? / ከሚከተሉት እኩልነቶች ውስጥ ትክክለኛው የትኛው ነው?

Options / አማራጮች:

- A. 1 L = 1 cm^3 [1 L = 1 cm^3]
- B. 1 mL = 1 cm^3 [1 mL = 1 cm^3]
- C. 1 mL = 1 dm^3 [1 mL = 1 dm^3]
- D. 1 L = 1 m^3 [1 L = 1 m^3]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): Since 1 dm^3 = 1000 cm^3 and 1 L = 1 dm^3, a liter equals 1000 cm^3, so one milliliter (1/1000 L) equals 1 cm^3. Option A is the main distractor, because a liter equals 1 dm^3 (1000 cm^3), not 1 cm^3.

ማብራሪያ (አማርኛ): 1 dm^3 = 1000 cm^3 እና 1 L = 1 dm^3 ስለሆነ አንድ ሊትር 1000 cm^3 ነው፤ ስለዚህ አንድ ሚሊሊትር (1/1000 L) ከ1 cm^3 ጋር እኩል ነው። አማራጭ A አሳሳች ነው፤ አንድ ሊትር 1 dm^3 (1000 cm^3) እንጂ 1 cm^3 አይደለም።

3.2 Density

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- **Definition:** The density of an object is its mass per unit volume.
- **Formula:** d = m / V
  - d is density
  - m is mass
  - V is volume
- **Example Calculation:**
  - An object has a mass of 15.0 g and a volume of 10.0 cm^3.
  - d = 15.0 g / 10.0 cm^3 = 1.50 g/cm^3
- **Importance:**
  - Density is an important **characteristic property** of a material.
  - It is independent of the quantity of material measured, as it is a ratio of mass to volume (an *intensive property*).
  - Can be used to determine the purity of a substance.
    - *Example:* An impure gold bar (mixed with silver or copper, which have lower densities) would be less dense than pure gold.

Example 2.1: Calculating the density of a substance

- **Problem:** Chaltu found a liquid's volume to be 35.1 mL and its mass to be 30.5 g. Identify the substance from the following list:
  - n-butyl alcohol: 0.810 g/mL
  - ethylene glycol: 1.114 g/mL
  - isopropyl alcohol: 0.785 g/mL
  - toluene: 0.866 g/mL
- **Solution:**
  - Calculate the density of the unknown liquid:
    - d = m / V = 30.5 g / 35.1 mL ≈ 0.869 g/mL
  - Compare with the table: The calculated density (0.869 g/mL) is very close to that of **toluene** (0.866 g/mL).
- **Answer Check:** Always report density in the units used for calculation (e.g., g/mL, g/cm^3, or g/L for gases).

📝 Practice Question 12 / የልምምድ ጥያቄ 12 (EUEE-style)

Question / ጥያቄ: Which statement about density is correct? / ስለ ጥግግት (density) የትኛው መግለጫ ትክክል ነው?

Options / አማራጮች:

- A. It is the mass per unit volume, it is an intensive property, and it can be used to judge the purity of a substance [በአንድ አሃድ መጠን ያለ ክብደት ነው፤ ኢንቴንሲቭ ባህሪ ነው፤ የንጥረ ነገርን ንጽሕናም ለመወሰን ይረዳል]
- B. It is the volume per unit mass [በአንድ አሃድ ክብደት ያለ መጠን ነው]
- C. It depends on the quantity of material measured [በሚለካው ንጥረ ነገር መጠን ላይ ይወሰናል]
- D. It is the force per unit area [በአንድ አሃድ ስፋት ላይ የሚሠራ ኃይል ነው]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): Density (d = m/V) is mass per unit volume; as a ratio it is independent of the quantity of material, so it is an intensive property and can indicate purity. Option C is the main distractor, because density does not change with the amount of material measured.

ማብራሪያ (አማርኛ): ጥግግት (d = m/V) በአንድ አሃድ መጠን ያለ ክብደት ነው፤ ጥምርታ በመሆኑ በንጥረ ነገሩ መጠን ላይ አይወሰንም፤ ስለዚህ ኢንቴንሲቭ ባህሪ ሲሆን ንጽሕናን ለማመልከትም ይረዳል። አማራጭ C አሳሳች ነው፤ ጥግግት በሚለካው ንጥረ ነገር መጠን አይለወጥም።

📝 Practice Question 13 / የልምምድ ጥያቄ 13 (EUEE-style)

Question / ጥያቄ: An object has a mass of 24.0 g and a volume of 8.0 cm^3. What is its density? / አንድ ነገር 24.0 g ክብደትና 8.0 cm^3 መጠን አለው። ጥግግቱ ስንት ነው?

Options / አማራጮች:

- A. 0.33 g/cm^3 [0.33 g/cm^3]
- B. 32 g/cm^3 [32 g/cm^3]
- C. 192 g/cm^3 [192 g/cm^3]
- D. 3.0 g/cm^3 [3.0 g/cm^3]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): d = m / V = 24.0 g / 8.0 cm^3 = 3.0 g/cm^3, reported with two significant figures because 8.0 has two. Option A is the main distractor, because it inverts the formula (V / m).

ማብራሪያ (አማርኛ): d = m / V = 24.0 g / 8.0 cm^3 = 3.0 g/cm^3 ነው፤ 8.0 ሁለት ጠቃሚ አሃዞች ስላሉት ውጤቱ በሁለት ጠቃሚ አሃዞች ይጻፋል። አማራጭ A ቀመሩን (V / m) ስለሚገለብጥ አሳሳች ነው።

📝 Practice Question 14 / የልምምድ ጥያቄ 14 (EUEE-style)

Question / ጥያቄ: A student finds that a liquid has a volume of 25.0 mL and a mass of 27.9 g. Using the densities in the guide's list, which substance is it most likely to be? / አንድ ተማሪ የአንድ ፈሳሽ መጠን 25.0 mL እና ክብደቱ 27.9 g መሆኑን አገኘ። በመመሪያው ዝርዝር ውስጥ ያሉትን ጥግግቶች በመጠቀም ፈሳሹ ምን የመሆን ዕድሉ ከፍተኛ ነው?

Options / አማራጮች:

- A. n-butyl alcohol (0.810 g/mL) [ኤን-ቡቲል አልኮሆል (0.810 g/mL)]
- B. isopropyl alcohol (0.785 g/mL) [አይሶፕሮፒል አልኮሆል (0.785 g/mL)]
- C. ethylene glycol (1.114 g/mL) [ኤቲሊን ግላይኮል (1.114 g/mL)]
- D. toluene (0.866 g/mL) [ቶሉኢን (0.866 g/mL)]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): d = m / V = 27.9 g / 25.0 mL ≈ 1.12 g/mL, which is very close to ethylene glycol (1.114 g/mL). Toluene is the main distractor: it is the answer in the guide's own example (0.869 g/mL for 30.5 g in 35.1 mL), but its density of 0.866 g/mL is far from 1.12 g/mL.

ማብራሪያ (አማርኛ): d = m / V = 27.9 g / 25.0 mL ≈ 1.12 g/mL ሲሆን ይህ ከኤቲሊን ግላይኮል (1.114 g/mL) ጋር በጣም ቅርብ ነው። ቶሉኢን አሳሳች አማራጭ ነው፤ በመመሪያው ምሳሌ (30.5 g በ35.1 mL = 0.869 g/mL) መልሱ እሱ ነው፤ ነገር ግን ጥግግቱ 0.866 g/mL ከ1.12 g/mL በጣም የራቀ ነው።

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4. Common Prefixes Used in SI Units

SI units are modified decimally by a series of prefixes to represent very large or very small quantities. These prefixes are widely used and easily added to base units.

- **Prefix Table:**

Factor	Prefix	Symbol	Decimal Number	Example
10^{12}	Tera	T	1,000,000,000,000	1 Terameter (Tm) = 1 × 10^{12} m
10^{9}	Giga	G	1,000,000,000	1 Gigameter (Gm) = 1 × 10^{9} m
10^{6}	Mega	M	1,000,000	1 Megameter (Mm) = 1 × 10^{6} m

10^{3}	Kilo	k	1,000	1 kilometer (km) = 1 × 10^{3} m
10^{2}	Hecto	h	100	1 hectometer (hm) = 1 × 10^{2} m
10^{1}	Deca	da	10	1 decameter (dam) = 1 × 10^{1} m
10^{-1}	Deci	d	0.1	1 decimeter (dm) = 1 × 10^{-1} m
10^{-2}	Centi	c	0.01	1 centimeter (cm) = 1 × 10^{-2} m
10^{-3}	Milli	m	0.001	1 millimeter (mm) = 1 × 10^{-3} m
10^{-6}	Micro	μ	0.000 001	1 micrometer (μm) = 1 × 10^{-6} m
10^{-9}	Nano	n	0.000 000 001	1 nanometer (nm) = 1 × 10^{-9} m
10^{-12}	Pico	p	0.000 000 000 001	1 picometer (pm) = 1 × 10^{-12} m

- **Examples:**
  - 1 Megawatt = 1,000,000 watts
  - 1 kilogram = 1,000 grams
  - 1 μF = 1 microFarad = 1/1,000,000 Farad
- **Abbreviations/Symbols:** kV for kilovolts, kW for kilowatts, km for kilometer.

📝 Practice Question 15 / የልምምድ ጥያቄ 15 (EUEE-style)

Question / ጥያቄ: Which factor does the prefix *nano* represent? / ናኖ የሚለው ቅድመ ቅጥያ የትኛውን ምክንያት (factor) ይወክላል?

Options / አማራጮች:

- A. 10^{-3} [10^{-3}]
- B. 10^{-6} [10^{-6}]
- C. 10^{-12} [10^{-12}]
- D. 10^{-9} [10^{-9}]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Nano stands for 10^{-9} (0.000 000 001), so 1 nanometer (nm) = 1 × 10^{-9} m. Option B is the main distractor: 10^{-6} is the factor for micro (μ), not nano.

ማብራሪያ (አማርኛ): ናኖ 10^{-9} (0.000 000 001) ማለት ነው፤ ስለዚህ 1 ናኖሜትር (nm) = 1 × 10^{-9} m። አማራጭ B አሳሳች ነው፤ 10^{-6} የማይክሮ (μ) እንጂ የናኖ ምክንያት አይደለም።

📝 Practice Question 16 / የልምምድ ጥያቄ 16 (EUEE-style)

Question / ጥያቄ: Which prefix–factor pair is **NOT** correct? / የትኛው የቅድመ ቅጥያና የምክንያት ጥንድ ትክክል አይደለም?

Options / አማራጮች:

- A. Mega – 10^{6} [ሜጋ – 10^{6}]

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- B. Centi – 10^{-2} [ሴንቲ – 10^{-2}]
- C. Micro – 10^{-3} [ማይክሮ – 10^{-3}]
- D. Pico – 10^{-12} [ፒኮ – 10^{-12}]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Micro is 10^{-6} (0.000 001); the factor 10^{-3} belongs to milli. Options A, B and D match the prefix table in the guide.

ማብራሪያ (አማርኛ): ማይክሮ 10^{-6} (0.000 001) ሲሆን 10^{-3} የሚሊ ምክንያት ነው። አማራጮች A፣ B እና D በመመሪያው የቅድመ ቅጥያ ሰንጠረዥ መሠረት ትክክል ናቸው።

5. Uncertainty in Measurements

All measurements inherently involve some degree of uncertainty. It is often impossible to obtain the exact value of a quantity.

- **Uncertainty as an Interval:** The uncertainty defines an interval within which the result is expected to be found with a given probability. All values within this interval are considered equally probable to represent the true result.
- **Significance of Significant Figures:** It is crucial to indicate the margin of error in a measurement by clearly stating the **number of significant figures**.
  - The **last digit** in a significant figure representation is always understood to be **uncertain**.
  - *Example 1:* If a volume is 6 mL (uncertainty of 1 mL), the actual volume is between 5 mL and 7 mL. Here, the digit '6' is the only significant figure and is uncertain by ±1 mL.
  - *Example 2:* Using a finer device, if the volume is 6.0 mL (uncertainty of 0.1 mL), the actual value is between 5.9 mL and 6.1 mL. This indicates greater accuracy.
  - The amount of uncertainty depends on the measuring device and the user's ability.

5.1 Categories of Uncertainty

1. **Systematic Uncertainties:**
   - **Definition:** Consistently cause the measured value to be either too large or too small.
   - **Examples:** Reaction time, inaccurate meter sticks, optical parallax (apparent shift in position due to viewing angle), miscalibrated balances.
   - **Elimination:** In principle, systematic uncertainties can be eliminated if their existence is known and accounted for.
2. **Random Uncertainties:**
   - **Definition:** Variations in measurements that occur without a predictable pattern.
   - **Source:** For precise measurements, these arise from the estimated part of the measurement.
   - **Reduction:** Random uncertainty can be reduced by taking multiple measurements, but it can *never* be completely eliminated.

📝 Practice Question 17 / የልምምድ ጥያቄ 17 (EUEE-style)

Question / ጥያቄ: A volume is recorded as 6 mL with an uncertainty of 1 mL. Between which values does the actual volume lie? / አንድ መጠን 6 mL ተብሎ ከ1 mL እርግጠኛ አለመሆን ጋር ተመዝግቧል። ትክክለኛው መጠን በየትኞቹ እሴቶች መካከል ይገኛል?

Options / አማራጮች:

- A. Between 5 mL and 7 mL [በ5 mL እና 7 mL መካከል]
- B. Between 6 mL and 7 mL [በ6 mL እና 7 mL መካከል]
- C. Between 5.9 mL and 6.1 mL [በ5.9 mL እና 6.1 mL መካከል]
- D. Exactly 6 mL [በትክክል 6 mL]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): The uncertainty defines an interval, so 6 mL ± 1 mL means the actual volume is between 5 mL and 7 mL. Option C is the main distractor: that interval (5.9 to 6.1 mL) belongs to a finer device reading 6.0 mL with an uncertainty of 0.1 mL.

ማብራሪያ (አማርኛ): እርግጠኛ አለመሆን የእሴቶችን ክልል ይወስናል፤ ስለዚህ 6 mL ± 1 mL ማለት ትክክለኛው መጠን በ5 mL እና 7 mL መካከል ነው። አማራጭ C አሳሳች ነው፤ ያ ክልል (5.9 እስከ 6.1 mL) 6.0 mL ለሚያነብና 0.1 mL እርግጠኛ አለመሆን ላለው የተሻለ መሣሪያ ነው።

📝 Practice Question 18 / የልምምድ ጥያቄ 18 (EUEE-style)

Question / ጥያቄ: Which statement about categories of uncertainty is correct? / ስለ እርግጠኛ ያለመሆን ምድቦች የትኛው መግለጫ ትክክል ነው?

Options / አማራጮች:

- A. Random uncertainties can be completely eliminated by repeating measurements [የዘፈቀደ እርግጠኛ ያለመሆኖች መለኪያዎችን በመድገም ሙሉ በሙሉ ሊወገዱ ይችላሉ]
- B. Systematic uncertainties consistently make the measured value too large or too small, and can in principle be eliminated if they are known and accounted for [ስልታዊ እርግጠኛ ያለመሆኖች የተለካውን ዋጋ ሁልጊዜ በጣም ትልቅ ወይም በጣም ትንሽ ያደርጋሉ፤ ከታወቁና ከተካተቱ በመርህ ደረጃ ሊወገዱ ይችላሉ]
- C. Optical parallax is an example of a random uncertainty [የዓይን አቅጣጫ ልዩነት (parallax) የዘፈቀደ እርግጠኛ ያለመሆን ምሳሌ ነው]
- D. Systematic uncertainties are variations that occur without a predictable pattern [ስልታዊ እርግጠኛ ያለመሆኖች ሊተነበይ በሚችል ንድፍ የማይከሰቱ ልዩነቶች ናቸው]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): Systematic uncertainties push results consistently in one direction (e.g., miscalibrated balances) and can in principle be removed once known. Option A is the main distractor: random uncertainty can be reduced by taking multiple measurements but can never be completely eliminated.

ማብራሪያ (አማርኛ): ስልታዊ እርግጠኛ ያለመሆኖች ውጤቱን ሁልጊዜ በአንድ አቅጣጫ ይገፋሉ (ለምሳሌ በትክክል ያልተስተካከሉ ሚዛኖች)፤ ከታወቁ በመርህ ደረጃ ሊወገዱ ይችላሉ። አማራጭ A አሳሳች ነው፤ የዘፈቀደ እርግጠኛ አለመሆን በበርካታ መለኪያዎች ሊቀንስ ይችላል እንጂ ሙሉ በሙሉ ሊወገድ አይችልም።

5.2 Calculating Uncertainties

- Estimating uncertainty, especially for the mean value, is important.
- **Methods for Estimating Uncertainty in Mean Value:**
  - **a. Using the Half Range:**
    - **Procedure:** Calculate the range of readings (maximum – minimum), then divide by two.
    - **Example 2.2:**
      - Readings for diameter: 66 mm, 65 mm, 61 mm. Mean = 64 mm.
      - Range = 66 mm – 61 mm = 5 mm.
      - Uncertainty = (5 mm) / 2 = 2.5 mm.
      - **Result:** Diameter = 64 mm ± 2.5 mm.
      - *Note:* This is acceptable for a ruler graduated in mm (read to ± 0.5 mm).
  - **b. Using the Reading Furthest from the Mean:**
    - **Procedure:** Calculate the absolute difference between the mean and the reading that is furthest from it.
    - **Example 2.2:**
      - Mean = 64 mm. Furthest reading is 61 mm.

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        - Uncertainty = 64 mm – 61 mm = 3 mm.
        - **Result:** Diameter = 64 mm ± 3 mm.
    - **c. Using the Resolution of the Instrument:**
        - **When to use:** If only a single reading is taken, or if repeated readings have the same value. This accounts for the instrument's limitations.
        - **Procedure:** Uncertainty is often half the smallest division or resolution of the instrument.
        - **Example 1 (Ruler):** If three readings were all 64 mm, a meter rule (readable to 0.5 mm) means the actual value is between 63.5 mm and 64.5 mm.
            - Uncertainty = 0.5 mm.
            - **Result:** Diameter = 64 mm ± 0.5 mm.
        - **Example 2 (Digital Instrument):** An ammeter recording currents to 0.1 A. A displayed value of 0.4 A (could be 0.36 A or 0.44 A).
            - Resolution = 0.1 A.
            - Uncertainty in a reading = 0.05 A.
        - **Example 3 (Top-loading balance):** Typical uncertainty is 0.05 g.
            - A weighing of 23.25 g is reported as 23.25 g ± 0.05 g.
            - This means the correct reading is between 23.20 g and 23.30 g.
- **Expressing Uncertainty:**
    - **a. Absolute Uncertainty:** Expressed as measurement ± absolute uncertainty.
        - *Example:* 23.25 g ± 0.05 g
    - **b. Percent Uncertainty:**
        - **Formula:** % uncertainty = (absolute uncertainty / measurement) × 100
        - **Example:** For 23.25 g ± 0.05 g:
            - % uncertainty = (0.05 g / 23.25 g) × 100 ≈ 0.2%
        - **Result:** 23.25 g ± 0.2%

📝 Practice Question 19 / የልምምድ ጥያቄ 19 (EUEE-style)

Question / ጥያቄ: A digital ammeter records currents to a resolution of 0.1 A. What is the uncertainty in a single reading? / አንድ ዲጂታል አምሜትር ጅረቶችን በ0.1 A ጥራት ይመዘግባል። በአንድ ንባብ ውስጥ ያለው እርግጠኛ አለመሆን ስንት ነው?

Options / አማራጮች:

- A. 0.1 A [0.1 A]
- B. 0.5 A [0.5 A]
- C. 0.05 A [0.05 A]
- D. 0.01 A [0.01 A]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): The uncertainty is often half the smallest division or resolution of the instrument, so for a resolution of 0.1 A it is 0.05 A. Option A is the main distractor, because it uses the full resolution instead of half of it.

ማብራሪያ (አማርኛ): እርግጠኛ አለመሆን ብዙ ጊዜ የመሣሪያው ትንሹ ክፍል ወይም ጥራት ግማሽ ነው፤ ስለዚህ ለ0.1 A ጥራት 0.05 A ነው። አማራጭ A ግማሹን ሳይሆን ሙሉውን ጥራት ስለሚጠቀም አሳሳች ነው።

📝 Practice Question 20 / የልምምድ ጥያቄ 20 (EUEE-style)

Question / ጥያቄ: A mass is reported as 50.0 g ± 0.5 g. What is the percent uncertainty? / አንድ ክብደት 50.0 g ± 0.5 g ተብሎ ተመዝግቧል። የመቶኛ እርግጠኛ አለመሆኑ ስንት ነው?

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Options / አማራጮች:

- A. 0.1% [0.1%]
- B. 10% [10%]
- C. 0.01% [0.01%]
- D. 1.0% [1.0%]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): % uncertainty = (absolute uncertainty / measurement) × 100 = (0.5 g / 50.0 g) × 100 = 1.0%. Option B (10%) is the main distractor: it results from dividing 0.5 g by 5.0 g instead of by 50.0 g.

ማብራሪያ (አማርኛ): የመቶኛ እርግጠኛ አለመሆን = (ፍጹም እርግጠኛ አለመሆን / የተለካው ዋጋ) × 100 = (0.5 g / 50.0 g) × 100 = 1.0% ነው። አማራጭ B (10%) አሳሳች ነው፤ ይህ የሚገኘው 0.5 gን በ50.0 g ፋንታ በ5.0 g በማካፈል ነው።

📝 Practice Question 21 / የልምምድ ጥያቄ 21 (EUEE-style)

Question / ጥያቄ: Three readings of a diameter are 30 mm, 32 mm and 37 mm (mean = 33 mm). Using the method of the reading furthest from the mean, what is the uncertainty? / የአንድ ዲያሜትር ሦስት ንባቦች 30 mm፣ 32 mm እና 37 mm ናቸው (አማካይ = 33 mm)። ከአማካዩ በጣም የራቀውን ንባብ ዘዴ በመጠቀም እርግጠኛ አለመሆኑ ስንት ነው?

Options / አማራጮች:

- A. ± 3 mm [± 3 mm]
- B. ± 4 mm [± 4 mm]
- C. ± 3.5 mm [± 3.5 mm]
- D. ± 7 mm [± 7 mm]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): The reading furthest from the mean is 37 mm, and the absolute difference is 37 − 33 = 4 mm. Option C is the main distractor: 3.5 mm is the half-range result, (37 − 30) / 2, which is a different method.

ማብራሪያ (አማርኛ): ከአማካዩ በጣም የራቀው ንባብ 37 mm ሲሆን ፍጹም ልዩነቱ 37 − 33 = 4 mm ነው። አማራጭ C አሳሳች ነው፤ 3.5 mm የግማሽ ክልል ዘዴ ውጤት ነው፤ (37 − 30) / 2፣ ይህም የተለየ ዘዴ ነው።

6. Precision and Accuracy

These terms describe the quality of measurements, which are always subject to experimental error.

- **Precision:**
  - **Definition:** Refers to the *closeness of a set of values* obtained from identical measurements of a quantity.
  - *Characteristic:* Reproducibility of results. If you repeat a measurement multiple times, precise measurements will be very close to each other.
- **Accuracy:**
  - **Definition:** Refers to the *closeness of a single measurement to its true value*.
- **Goal of Scientists:** To obtain measured values that are **both accurate and precise**.

Example 2.4: Precision and Accuracy

- *Figure 2.5:* Uses archery targets to illustrate the concepts:
  - **a. Precise but Inaccurate:** All darts are clustered closely together, but far from the bullseye (true value).
  - **b. Accurate but Imprecise:** Darts are spread out, but their average position is near the bullseye.
  - **c. Both Precise and Accurate:** All darts are clustered closely together *and* around the bullseye.
  - **d. Neither Precise nor Accurate:** Darts are spread out *and* far from the bullseye.

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📝 Practice Question 22 / የልምምድ ጥያቄ 22 (EUEE-style)

Question / ጥያቄ: Which statement correctly defines *precision*? / የትኛው መግለጫ ትክክለኛ ተደጋጋሚነትን *(precision)* በትክክል ይተረጉማል?

Options / አማራጮች:

- A. The closeness of a set of values obtained from identical measurements of a quantity [ከአንድ ብዛት ተመሳሳይ መለኪያዎች የተገኙ የእሴቶች ስብስብ እርስ በርስ ያለው ቅርበት]
- B. The closeness of a single measurement to its true value [የአንድ መለኪያ ወደ እውነተኛ ዋጋው ያለው ቅርበት]
- C. The number of readings taken in an experiment [በሙከራ ውስጥ የተወሰዱ ንባቦች ብዛት]
- D. The difference between the largest and smallest unit in a system [በሥርዓት ውስጥ ባሉ ትልቁና ትንሹ አሃዶች መካከል ያለ ልዩነት]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): Precision is the closeness of repeated measurements to each other, that is, the reproducibility of results. Option B is the main distractor: it defines accuracy, the closeness of a single measurement to its true value.

ማብራሪያ (አማርኛ): ትክክለኛ ተደጋጋሚነት (precision) የተደጋገሙ መለኪያዎች እርስ በርስ ያላቸው ቅርበት ነው፣ ማለትም የውጤቶች ተደጋጋሚነት። አማራጭ B አሳሳች ነው፤ ይህ የትክክለኝነት (accuracy) ትርጉም ነው፣ የአንድ መለኪያ ወደ እውነተኛ ዋጋው ያለው ቅርበት።

📝 Practice Question 23 / የልምምድ ጥያቄ 23 (EUEE-style)

Question / ጥያቄ: On an archery target the darts are spread out, but their average position is near the bullseye. How is this described? / በቀስት ማነጣጠሪያ ላይ ፍላጻዎቹ ተበታትነዋል፤ ነገር ግን አማካይ ቦታቸው ወደ ማዕከሉ ቅርብ ነው። ይህ እንዴት ይገለጻል?

Options / አማራጮች:

- A. Precise but inaccurate [ትክክለኛ ተደጋጋሚ ግን ትክክል ያልሆነ]
- B. Both precise and accurate [ትክክለኛ ተደጋጋሚም ትክክልም]
- C. Neither precise nor accurate [ትክክለኛ ተደጋጋሚም ትክክልም ያልሆነ]
- D. Accurate but imprecise [ትክክል ግን ትክክለኛ ተደጋጋሚ ያልሆነ]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Darts that are spread out are imprecise, but because their average is near the bullseye the result is accurate. Option A is the main distractor, because it describes darts clustered together but far from the bullseye.

ማብራሪያ (አማርኛ): የተበታተኑ ፍላጻዎች ትክክለኛ ተደጋጋሚ አይደሉም፤ ነገር ግን አማካያቸው ወደ ማዕከሉ ቅርብ ስለሆነ ውጤቱ ትክክል ነው። አማራጭ A አሳሳች ነው፤ ይህ የተሰባሰቡ ግን ከማዕከሉ የራቁ ፍላጻዎችን ይገልጻል።

📝 Practice Question 24 / የልምምድ ጥያቄ 24 (EUEE-style)

Question / ጥያቄ: The true mass of an object is 3.00 g. A student measures it three times and gets 2.50 g, 2.51 g and 2.49 g. How can these measurements be described? / የአንድ ነገር እውነተኛ ክብደት 3.00 g ነው። ተማሪው ሦስት ጊዜ ሲለካ 2.50 g፣ 2.51 g እና 2.49 g አገኘ። እነዚህ መለኪያዎች እንዴት ይገለጻሉ?

Options / አማራጮች:

- A. Accurate but imprecise [ትክክል ግን ትክክለኛ ተደጋጋሚ ያልሆኑ]
- B. Precise but inaccurate [ትክክለኛ ተደጋጋሚ ግን ትክክል ያልሆኑ]
- C. Both precise and accurate [ትክክለኛ ተደጋጋሚም ትክክልም]
- D. Neither precise nor accurate [ትክክለኛ ተደጋጋሚም ትክክልም ያልሆኑ]

Correct Answer / ትክክለኛ መልስ: B

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Explanation (English): The three readings are very close to each other, so they are precise, but they are far from the true value of 3.00 g, so they are not accurate. Option C is the main distractor, since closeness among the readings alone does not make them accurate.

ማብራሪያ (አማርኛ): ሦስቱ ንባቦች እርስ በርስ በጣም ቅርብ ስለሆኑ ትክክለኛ ተደጋጋሚ ናቸው፤ ነገር ግን ከእውነተኛው ዋጋ 3.00 g ስለሚርቁ ትክክል አይደሉም። አማራጭ C አሳሳች ነው፤ ንባቦቹ እርስ በርስ መቀራረባቸው ብቻ ትክክል አያደርጋቸውም።

7. Significant Figures

Significant figures (also called meaningful digits) are crucial for indicating the exactness (uncertainty) of a measured or calculated quantity. The last digit in a significant figure is always uncertain.

7.1 Rules for Determining Significant Figures

1. **Any digit that is not zero is significant.**
   - *Examples:* 845 cm (3 significant figures), 1.234 kg (4 significant figures).
2. **Zeros between nonzero digits are significant.**
   - *Examples:* 606 m (3 significant figures), 40,501 kg (5 significant figures).
3. **Zeros to the left of the first nonzero digit are not significant.** Their purpose is only to indicate the decimal point's placement.
   - *Examples:* 0.08 L (1 significant figure), 0.0000349 g (3 significant figures).
4. **Trailing Zeros (zeros at the end of a number):**
   - **If a number is greater than 1:** All zeros written to the right of the decimal point count as significant figures.
     - *Examples:* 2.0 mg (2 significant figures), 40.062 mL (5 significant figures), 3.040 dm (4 significant figures).
   - **If a number is less than 1:** Only the zeros that are at the *end of the number* and the zeros that are *between nonzero digits* are significant.
     - *Examples:* 0.090 kg (2 significant figures), 0.3005 L (4 significant figures), 0.00420 min (3 significant figures).
5. **Numbers without decimal points (trailing zeros):** Trailing zeros may or may not be significant, leading to ambiguity.
   - *Example:* 400 cm could have 1 (4), 2 (40), or 3 (400) significant figures.
   - **Resolution:** Use **scientific notation** to avoid this ambiguity.
     - For 400:
       - 1 significant figure: 4×10^2
       - 2 significant figures: 4.0×10^2
       - 3 significant figures: 4.00×10^2

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Example 2.5: Significant Figures

- **Problem:** Determine the number of significant figures in: (a) 394 cm, (b) 5.03 g, (c) 0.714 m, (d) 0.052 kg, (e) 2.720×10^{22} atoms, (f) 3000 mL.
- **Solution:**
  - **(a) 394 cm:** Three (all nonzero digits are significant).
  - **(b) 5.03 g:** Three (zeros between nonzero digits are significant).
  - **(c) 0.714 m:** Three (leading zeros are not significant).
  - **(d) 0.052 kg:** Two (leading zeros are not significant).
  - **(e) 2.720×10^{22} atoms:** Four (number is greater than 1, so zeros to the right of the decimal are significant).
  - **(f) 3000 mL:** Ambiguous. Could be four (3.000×10^3), three (3.00×10^3), two (3.0×10^3), or one (3×10^3). Scientific notation is needed to clarify.

📝 Practice Question 25 / የልምምድ ጥያቄ 25 (EUEE-style)

Question / ጥያቄ: How many significant figures are in 606 m? / በ606 m ውስጥ ስንት ጠቃሚ አሃዞች (significant figures) አሉ?

Options / አማራጮች:

- A. 1 [1]
- B. 2 [2]
- C. 3 [3]
- D. 4 [4]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Zeros between nonzero digits are significant, so 6, 0 and 6 are all significant, giving three significant figures. Option B is the main distractor, because it wrongly ignores the zero as if it were a leading zero.

ማብራሪያ (አማርኛ): በዜሮ ያልሆኑ አሃዞች መካከል ያሉ ዜሮዎች ጠቃሚ ናቸው፤ ስለዚህ 6፣ 0 እና 6 ሁሉም ጠቃሚ ሲሆኑ ሦስት ጠቃሚ አሃዞች ይገኛሉ። አማራጭ B ዜሮውን እንደ መጀመሪያ ዜሮ በስህተት ስለሚተወው አሳሳች ነው።

📝 Practice Question 26 / የልምምድ ጥያቄ 26 (EUEE-style)

Question / ጥያቄ: How many significant figures are in 0.090 kg? / በ0.090 kg ውስጥ ስንት ጠቃሚ አሃዞች አሉ?

Options / አማራጮች:

- A. 2 [2]
- B. 3 [3]
- C. 1 [1]
- D. 4 [4]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): The zeros to the left of the first nonzero digit (9) are not significant, but the zero at the end of this number less than 1 is significant, so there are two significant figures (9 and 0). Option C is the main distractor, because it counts only the 9 and ignores the significant trailing zero.

ማብራሪያ (አማርኛ): ከመጀመሪያው ዜሮ ያልሆነ አሃዝ (9) በስተግራ ያሉ ዜሮዎች ጠቃሚ አይደሉም፤ ነገር ግን ከ1 በታች በሆነው በዚህ ቁጥር መጨረሻ ላይ ያለው ዜሮ ጠቃሚ ነው፤ ስለዚህ ሁለት ጠቃሚ አሃዞች (9 እና 0) አሉ። አማራጭ C 9ን ብቻ ቆጥሮ ጠቃሚውን የመጨረሻ ዜሮ ስለሚተው አሳሳች ነው።

📝 Practice Question 27 / የልምምድ ጥያቄ 27 (EUEE-style)

Question / ጥያቄ: Which of the following measurements has the greatest number of significant figures? / ከሚከተሉት መለኪያዎች ውስጥ ከፍተኛው የጠቃሚ አሃዞች ብዛት ያለው የትኛው ነው?

Options / አማራጮች:

- A. 3.040 dm [3.040 dm]
- B. 0.3005 L [0.3005 L]
- C. 0.00420 min [0.00420 min]
- D. 40,501 kg [40,501 kg]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): 40,501 kg has five significant figures (zeros between nonzero digits count), while 3.040 dm and 0.3005 L have four each and 0.00420 min has three. Options A and B are the main distractors, since they have four significant figures, one fewer than D.

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ማብራሪያ (አማርኛ)**:** 40,501 kg አምስት ጠቃሚ አሃዞች አሉት (በዜሮ ያልሆኑ አሃዞች መካከል ያሉ ዜሮዎች ይቆጠራሉ)፤ 3.040 dm እና 0.3005 L እያንዳንዳቸው አራት፣ 0.00420 min ደግሞ ሦስት አላቸው። አማራጮች A እና B ከD አንድ ያነሰ (አራት) ጠቃሚ አሃዞች ስላላቸው አሳሳች ናቸው።

7.2 Rules for Handling Significant Figures in Calculations

1. **Addition and Subtraction:**
    - The answer cannot have more digits to the right of the decimal point than *either of the original numbers* with the fewest decimal places.
    - **Rounding-off Procedure:**
        - If the first digit to be dropped is less than 5, simply drop it and all subsequent digits.
        - If the first digit to be dropped is 5 or greater, add 1 to the preceding digit.
    - *Examples:*

```
89.332
+ 1.1  <- one decimal place
------
90.432  -> round off to 90.4

2.097
- 0.12  <- two decimal places
------
1.977   -> round off to 1.98
```

2. **Multiplication and Division:**
    - The number of significant figures in the final product or quotient is determined by the original number that has the *smallest number of significant figures*.
    - *Examples:*
        - 2.8 × 4.5039 = 12.61092 → round off to 13 (2.8 has 2 sig figs)
        - 6.85 / 112.04 = 0.0611388789 → round off to 0.0611 (6.85 has 3 sig figs)
3. **Exact Numbers:**
    - Numbers obtained from definitions (e.g., 1 ft = 12 in, where 12 is exact) or by counting objects are considered to have an **infinite number of significant figures**. They do not limit the significant figures in a calculation.

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📝 Practice Question 28 / የልምምድ ጥያቄ 28 (EUEE-style)

Question / ጥያቄ: What is the sum 14.25 + 3.1 + 0.456 reported with the correct number of decimal places? / የ14.25 + 3.1 + 0.456 ድምር በትክክለኛ የአስርዮሽ ቦታዎች ብዛት ሲጻፍ ስንት ነው?

Options / አማራጮች:

- A. 17.806 [17.806]
- B. 17.8 [17.8]
- C. 17.81 [17.81]
- D. 18 [18]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): The raw sum is 17.806, and in addition the answer cannot have more decimal places than the number with the fewest, which is 3.1 with one decimal place, so the answer is 17.8. Option C is the main distractor, because it keeps two decimal places, the count of 14.25 rather than the smallest count.

ማብራሪያ (አማርኛ): ጥሬ ድምሩ 17.806 ነው፤ በመደመር ጊዜ መልሱ ከዝቅተኛው የአስርዮሽ ቦታ ካለው ቁጥር የበለጠ የአስርዮሽ ቦታ ሊኖረው አይችልም፤ ይህ 3.1 (አንድ የአስርዮሽ ቦታ) ነው፤ ስለዚህ መልሱ 17.8 ነው። አማራጭ C አሳሳች ነው፤ ዝቅተኛውን ሳይሆን የ14.25ን ሁለት የአስርዮሽ ቦታዎች ይይዛል።

📝 Practice Question 29 / የልምምድ ጥያቄ 29 (EUEE-style)

Question / ጥያቄ: Calculate 3.2 × 4.567 and report the answer with the correct number of significant figures. / 3.2 × 4.567ን አስልተው መልሱን በትክክለኛ የጠቃሚ አሃዞች ብዛት ይጻፉ።

Options / አማራጮች:

- A. 14.614 [14.614]
- B. 14.6 [14.6]
- C. 15 [15]
- D. 14.61 [14.61]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): The product is 14.6144, and in multiplication the answer has as many significant figures as the number with the fewest; 3.2 has two, so the answer rounds to 15. Option B (14.6) is the main distractor, because it keeps three significant figures, one more than 3.2 allows.

ማብራሪያ (አማርኛ): ብዜቱ 14.6144 ነው፤ በማባዛት ጊዜ መልሱ ዝቅተኛ ጠቃሚ አሃዞች ካለው ቁጥር ጋር እኩል ይሆናል፤ 3.2 ሁለት ስላለው መልሱ ወደ 15 ይጠጋጋል። አማራጭ B (14.6) አሳሳች ነው፤ 3.2 ከሚፈቅደው አንድ የበለጠ ሦስት ጠቃሚ አሃዞችን ይይዛል።

📝 Practice Question 30 / የልምምድ ጥያቄ 30 (EUEE-style)

Question / ጥያቄ: There are 12 test tubes (a counted number), and each has a mass of 1.25 g. What is the total mass reported with the correct number of significant figures? / 12 የሙከራ ቱቦዎች (የተቆጠረ ቁጥር) አሉ፤ እያንዳንዳቸው 1.25 g ክብደት አላቸው። ጠቅላላ ክብደቱ በትክክለኛ የጠቃሚ አሃዞች ብዛት ሲጻፍ ስንት ነው?

Options / አማራጮች:

- A. 15.0 g [15.0 g]
- B. 15 g [15 g]
- C. 15.00 g [15.00 g]
- D. 20 g [20 g]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): A counted number is exact and has infinite significant figures, so only 1.25 g (three significant figures) limits the product 12 × 1.25 g = 15.0 g. Option B is the main distractor, because it wrongly treats 12 as a two-significant-figure number.

ማብራሪያ (አማርኛ): የተቆጠረ ቁጥር ትክክለኛ ሲሆን ገደብ የሌለው የጠቃሚ አሃዞች ብዛት አለው፤ ስለዚህ ብዜቱን 12 × 1.25 g = 15.0 g የሚገድበው 1.25 g (ሦስት ጠቃሚ አሃዞች) ብቻ ነው። አማራጭ B 12ን ባለ ሁለት ጠቃሚ አሃዝ ቁጥር አድርጎ በስህተት ስለሚቆጥር አሳሳች ነው።

8. Scientific Notation and Decimal Places

Scientific notation (exponential notation) is used to conveniently write very large or very small numbers.

- **Format:** One nonzero digit is placed to the left of the decimal point.
  - The number is expressed as a $\times 10^n$, where ‘a’ is a number between 1 and 10 (inclusive of 1, exclusive of 10) and ‘n’ is an integer exponent.

- **Converting to Scientific Notation:**
  - **Large Numbers:** Move the decimal point to the left until one nonzero digit remains to its left. The exponent ‘n’ is the number of places the decimal was moved.
    - *Example:* 602,000,000,000,000,000,000,000 = 6.02×10^{23} (23 places to the left, so exponent is 23).
  - **Small Numbers:** Move the decimal point to the right until one nonzero digit remains to its left. The exponent ‘n’ is the negative of the number of places the decimal was moved.
    - *Example:* 0.000 000 000 000 000 000 000 327 = 3.27×10^{-22} (22 places to the right, so exponent is –22).
- **Converting from Scientific Notation to Decimal Form:** Reverse the process based on the exponent.

Example 2.6: Unit Conversions with Scientific Notation

- **Problem:** The radius of a phosphorus atom is 1.10 Å. Express this distance in centimeters (cm) and nanometers (nm).
  - Given: 1 Å = 1×10^{-10} m, 1 cm = 1×10^{-2} m, 1 nm = 1×10^{-9} m.
- **Plan:** Use unit factors from the given equalities to convert from Å to m, then to cm or nm.
- **Solution:**
  - **To Centimeters (cm):**
    - x cm = 1.10 Å × (1×10^{-10} m / 1 Å) × (1 cm / 1×10^{-2} m) = 1.10×10^{-8} cm
  - **To Nanometers (nm):**
    - y nm = 1.10 Å × (1×10^{-10} m / 1 Å) × (1 nm / 1×10^{-9} m) = 1.10×10^{-1} nm

📝 Practice Question 31 / የልምምድ ጥያቄ 31 (EUEE-style)

Question / ጥያቄ: Which of the following numbers is correctly written in scientific notation? / ከሚከተሉት ቁጥሮች ውስጥ በሳይንሳዊ አጻጻፍ (scientific notation) በትክክል የተጻፈው የትኛው ነው?

Options / አማራጮች:

- A. 12.5×10^{3} [12.5×10^{3}]
- B. 0.85×10^{-2} [0.85×10^{-2}]
- C. 4.0×10^{2} [4.0×10^{2}]
- D. 25×10^{-1} [25×10^{-1}]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): In scientific notation $a \times 10^{n}$, ‘a’ must be at least 1 and less than 10, and only 4.0 meets this rule. Option B is the main distractor, because 0.85 is less than 1, so it is not in correct scientific notation.

ማብራሪያ (አማርኛ): በሳይንሳዊ አጻጻፍ $a \times 10^{n}$ ውስጥ ‘a’ ከ1 ያላነሰና ከ10 ያነሰ መሆን አለበት፤ ይህን ሕግ የሚያሟላው 4.0 ብቻ ነው። አማራጭ B አሳሳች ነው፤ 0.85 ከ1 ያነሰ ስለሆነ በትክክለኛ ሳይንሳዊ አጻጻፍ አይደለም።

📝 Practice Question 32 / የልምምድ ጥያቄ 32 (EUEE-style)

Question / ጥያቄ: Given 1 Å = 1×10^{-10} m and 1 nm = 1×10^{-9} m, express a distance of 2.50 Å in nanometers. / 1 Å = 1×10^{-10} m እና 1 nm = 1×10^{-9} m ከተሰጠ 2.50 Å የሆነን ርቀት በናኖሜትር ይግለጹ።

Options / አማራጮች:

- A. 2.50×10^{-8} nm [2.50×10^{-8} nm]
- B. 2.50×10^{-1} nm [2.50×10^{-1} nm]
- C. 2.50×10^{1} nm [2.50×10^{1} nm]
- D. 2.50×10^{-10} nm [2.50×10^{-10} nm]

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Correct Answer / ትክክለኛ መልስ: B

Explanation (English): 2.50 Å × (1 × 10^{-10} m / 1 Å) × (1 nm / 1 × 10^{-9} m) = 2.50 × 10^{-1} nm, using the same unit-factor method as Example 2.6. Option A is the main distractor: 10^{-8} is the exponent for the answer in centimeters (as in 1.10 × 10^{-8} cm), not nanometers.

ማብራሪያ (አማርኛ): 2.50 Å × (1 × 10^{-10} m / 1 Å) × (1 nm / 1 × 10^{-9} m) = 2.50 × 10^{-1} nm ነው፤ ይህም በምሳሌ 2.6 ውስጥ ያለውን ተመሳሳይ የአሃድ ምክንያት ዘዴ ይጠቀማል። አማራጭ A አሳሳች ነው፤ 10^{-8} በሴንቲሜትር ለሚሰጥ መልስ የሚሆን ኤክስፖነንት ነው (እንደ 1.10 × 10^{-8} cm)፣ ለናኖሜትር አይደለም።

9. Chemistry as Experimental Science

Chemistry is fundamentally an experimental science, with much of its knowledge derived from laboratory research and practical investigation.

- **Experimental Foundation:**
  - All chemical statements are **evidence-based**, relying on experiments.
  - Chemistry shares the **experimental method** with all other sciences and continuously improves through new discoveries.
- **Modern Chemistry:** Today's chemists utilize:
  - Computers to study microscopic structures and chemical properties.
  - Sophisticated electronic equipment for analysis (e.g., pollutants, toxic substances).
- **Diverse Applications:** Chemistry is often called the "central science" due to its broad applications and connections to other fields.
  - **Biology and Medicine:** Development of new drugs, exploration of biological processes at atomic/molecular levels.
  - **Agriculture:** Agricultural research.
  - **Environmental Science:** Seeking solutions for pollution, developing energy replacements.
  - **Industry:** Basis for many products (e.g., polymers for clothing, utensils, artificial organs, toys).
- **Interdisciplinary Role:** Provides theoretical and experimental tools to other sciences like Biology and Geology, which cannot be studied without understanding chemical phenomena.

📝 Practice Question 33 / የልምምድ ጥያቄ 33 (EUEE-style)

Question / ጥያቄ: Why is chemistry described as an experimental science? / ኬሚስትሪ የሙከራ ሳይንስ (experimental science) ተብሎ የሚገለጸው ለምንድን ነው?

Options / አማራጮች:

- A. Because it is studied only with computers [በኮምፒውተሮች ብቻ ስለሚጠና]
- B. Because its facts are never tested [እውነታዎቹ ፈጽሞ ስለማይፈተኑ]
- C. Because it does not share its method with other sciences [ዘዴውን ከሌሎች ሳይንሶች ጋር ስለማይጋራ]
- D. Because all chemical statements are evidence-based and rely on experiments [ሁሉም የኬሚስትሪ መግለጫዎች በማስረጃ ላይ የተመሠረቱና በሙከራዎች ላይ የሚደገፉ ስለሆኑ]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): Chemical statements are evidence-based and rely on experiments, and chemistry shares the experimental method with all other sciences. Option A is the main distractor: modern chemists do use computers, but only to study microscopic structures and properties, alongside laboratory experiments.

ማብራሪያ (አማርኛ): የኬሚስትሪ መግለጫዎች በማስረጃ ላይ የተመሠረቱና በሙከራዎች የሚደገፉ ናቸው፤ ኬሚስትሪም የሙከራ ዘዴን ከሌሎች ሳይንሶች ሁሉ ጋር ይጋራል። አማራጭ A አሳሳች ነው፤ የዘመናችን ኬሚስቶች ኮምፒውተሮችን ይጠቀማሉ፤ ነገር ግን ከላቦራቶሪ ሙከራዎች ጋር በአጉሊ መነጽር ደረጃ ያሉ አወቃቀሮችንና ባህሪያትን ለማጥናት ብቻ ነው።

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10. The Scientific Method

The scientific method is a systematic process that scientists use to investigate, verify, or construct an accurate and reliable understanding of natural phenomena. It provides an objective framework for scientific inquiry and analysis.

- **Aim:** To analyze observations and arrive at conclusions that either support or contradict initial observations.
- **General Steps:** While variations exist, a generally accepted sequence of steps forms the scientific method.

10.1 Scientific Method Steps

1. **Observation and Formulation of a Question:**
    - **First step:** Making an observation about an aspect of the universe.
    - **Followed by:** Asking a question related to that observation.
    - *Examples:* "Why is the sky black at night?" or "Why is air invisible?"
2. **Data Collection and Hypothesis:**
    - **Data Collection:** Gathering all relevant information and data related to the observation.
    - **Hypothesis Formulation:** Developing a testable explanation or prediction based on the observation and collected data.
        - A hypothesis could describe the cause, effect, or relationship of phenomena.
3. **Testing the Hypothesis (Experimentation):**
    - **Purpose:** To scientifically test the hypothesis.
    - **Method:** Scientists conduct controlled experiments.
    - **Goal:** Determine whether the hypothesis agrees with or contradicts real-world observations.
    - **Outcome:** Experimental results increase or decrease confidence in the hypothesis.
4. **Analysis and Conclusion:**
    - **Analysis:** Using proper mathematical and scientific procedures to interpret the experimental results.
    - **Conclusion:**
        - If the data is consistent with the hypothesis, the hypothesis is accepted.
        - If the data contradicts the hypothesis, it is rejected or modified, and then reanalyzed.
    - **Repetition:** A hypothesis cannot be proven or disproven by a single experiment. Repeated experimentation is necessary to eliminate discrepancies.
    - **Theory:** When a hypothesis is repeatedly tested and proven beyond reasonable doubt with no discrepancies, it can be accepted as a theory.

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📝 Practice Question 34 / የልምምድ ጥያቄ 34 (EUEE-style)

Question / ጥያቄ: Which option gives the steps of the scientific method in the correct order? / የትኛው አማራጭ የሳይንሳዊ ዘዴ ደረጃዎችን በትክክለኛ ቅደም ተከተል ይሰጣል?

Options / አማራጮች:

- A. Observation and question → data collection and hypothesis → testing the hypothesis → analysis and conclusion [ምልከታና ጥያቄ → መረጃ መሰብሰብና መላምት → መላምትን መፈተሽ → ትንተናና መደምደሚያ]
- B. Hypothesis → observation and question → analysis and conclusion → testing [መላምት → ምልከታና ጥያቄ → ትንተናና መደምደሚያ → ፍተሻ]
- C. Conclusion → testing the hypothesis → observation and question → hypothesis [መደምደሚያ → መላምትን መፈተሽ → ምልከታና ጥያቄ → መላምት]
- D. Testing the hypothesis → data collection → conclusion → observation [መላምትን መፈተሽ → መረጃ መሰብሰብ → መደምደሚያ → ምልከታ]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): The guide's sequence is: observation and formulation of a question, data collection and hypothesis, testing the hypothesis through experiments, and finally analysis and conclusion. Option B is the main distractor, because it starts with the hypothesis before any observation or data has been gathered.

ማብራሪያ (አማርኛ): በመመሪያው ቅደም ተከተል መሠረት፦ ምልከታና ጥያቄ ማቅረብ፣ መረጃ መሰብሰብና መላምት ማዘጋጀት፣ መላምቱን በሙከራ መፈተሽ፣ በመጨረሻም ትንተናና መደምደሚያ ናቸው። አማራጭ B አሳሳች ነው፤ ምንም ምልከታ ወይም መረጃ ሳይሰበሰብ በመላምት ይጀምራል።

📝 Practice Question 35 / የልምምድ ጥያቄ 35 (EUEE-style)

Question / ጥያቄ: Which statement about hypotheses and theories is correct? / ስለ መላምቶችና ንድፈ ሐሳቦች (theories) የትኛው መግለጫ ትክክል ነው?

Options / አማራጮች:

- A. A single experiment is enough to prove a hypothesis [አንድ ሙከራ መላምትን ለማረጋገጥ በቂ ነው]
- B. A hypothesis that the data contradicts is always accepted unchanged [መረጃው የሚቃረነው መላምት ሁልጊዜ ሳይቀየር ተቀባይነት ያገኛል]
- C. A hypothesis cannot be proven or disproven by a single experiment, and one that is repeatedly tested with no discrepancies can be accepted as a theory [መላምት በአንድ ሙከራ ሊረጋገጥም ሊተባበልም አይችልም፤ በተደጋጋሚ ተፈትኖ ምንም ልዩነት የሌለው መላምት እንደ ንድፈ ሐሳብ ተቀባይነት ሊያገኝ ይችላል]
- D. A theory is a hypothesis that has never been tested [ንድፈ ሐሳብ ፈጽሞ ያልተፈተነ መላምት ነው]

Correct Answer / ትክክለኛ መልስ: C

Explanation (English): Repeated experimentation is needed to eliminate discrepancies, and a hypothesis tested repeatedly beyond reasonable doubt can be accepted as a theory. Option D is the main distractor, because a theory comes from repeated testing, not from a lack of testing.

ማብራሪያ (አማርኛ): ልዩነቶችን ለማስወገድ ተደጋጋሚ ሙከራ ያስፈልጋል፤ ያለ ምንም ጥርጣሬ በተደጋጋሚ የተፈተነ መላምት እንደ ንድፈ ሐሳብ ተቀባይነት ሊያገኝ ይችላል። አማራጭ D አሳሳች ነው፤ ንድፈ ሐሳብ የሚመጣው ከተደጋጋሚ ፍተሻ እንጂ ከፍተሻ እጦት አይደለም።

11. Some Experimental Skills in Chemistry

11.1 Laboratory Safety Rules

Laboratory safety is paramount. Disregarding safety protocols endangers oneself and others. Adhering to all safety rules is the best practice.

1. Responsible behavior is essential; thoughtless actions creating dangers (spills, broken glass) are unacceptable.
2. Wear approved eye protection at all times in the laboratory and chemical handling areas, unless explicitly instructed otherwise.
3. Perform no unauthorized experiments, including using only specified quantities. Consult your teacher for any doubts.
4. Do not smoke in the laboratory; it's a fire hazard and draws chemicals into lungs.
5. In case of fire or accident, call the teacher at once. Know the location of fire extinguishers and safety showers.
6. Report all injuries to your instructor at once. Medical treatment may be required for cuts, burns, or fume inhalation.
7. Do not eat or drink anything in the laboratory.
8. Avoid breathing fumes of any kind.
9. Never use mouth suction in filling pipets. Always use a suction device.
10. Never work alone in the laboratory. At least one other person and your teacher must be quickly available.
11. Wear shoes in the laboratory. Bare feet and sandals are prohibited due to broken glass and chemical spill hazards.
12. Confine long hair and loose clothing (e.g., ties). They can catch fire or become chemically contaminated.

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- A laboratory apron or lab coat provides protection and is required for easily combustible clothing.
- Wearing old clothing is advisable as it is often less loose/flammable and less expensive to replace.

13. Keep your work area neat at all times. Clean up spills and broken glass immediately. Clean your workspace and put away equipment at the end of the period.
14. Be careful when heating liquids. Add boiling chips to avoid “bumping.” Flammable liquids (ethers, hydrocarbons, alcohols, acetone, carbon disulfide) must never be heated over an open flame.
15. Always pour acids into water when mixing. This prevents violent spattering.
16. Do not force a rubber stopper onto glass tubing or thermometers. Lubricate with glycerol or water. Use paper/cloth to protect hands. Grasp glass close to the stopper.
17. Dispose of excess liquid reagents by flushing small quantities down the sink. Consult the teacher for large quantities. Dispose of solids in crocks. Never return reagents to the dispensing bottle.
18. Carefully read the experiment and answer prelab questions before coming to the laboratory. An unprepared student is a hazard.
19. Test tubes being heated or containing reacting mixtures should never be pointed at anyone. Speak to neighbors or the teacher if this practice is observed.
20. If you have a cut on your hand, cover it with a bandage or wear appropriate laboratory gloves.
21. Think about what you are doing. Plan ahead and avoid just following instructions blindly, as this predisposes you to accidents.

Key Safety Equipment:

- **Safety goggles:** Required in all labs to protect eyes from splashes.
- **Latex gloves:** Should be used when corrosive chemicals might spill on hands.
- **Lab apron or coat:** Prevents injury from spills or splashes.

📝 Practice Question 36 / የልምምድ ጥያቄ 36 (EUEE-style)

Question / ጥያቄ: Which is the correct and safe way to mix an acid with water in the laboratory? / በላቦራቶሪ ውስጥ አሲድን ከውሃ ጋር ለመቀላቀል ትክክለኛውና ደኅንነቱ የተጠበቀው መንገድ የትኛው ነው?

Options / አማራጮች:

- A. Always pour water into the acid [ሁልጊዜ ውሃውን በአሲዱ ላይ ማፍሰስ]
- B. Always pour the acid into water, which prevents violent spattering [ሁልጊዜ አሲዱን በውሃ ውስጥ ማፍሰስ፤ ይህም የኃይለኛ ርጭትን ይከላከላል]
- C. The order does not matter [ቅደም ተከተሉ ምንም አይደለም]
- D. Mix them over an open flame [በክፍት እሳት ላይ ማቀላቀል]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): The safety rules say to always pour acids into water when mixing, because this prevents violent spattering. Option A is the main distractor, since it reverses the safe order.

ማብራሪያ (አማርኛ): የደኅንነት ሕጎች ሲቀላቀሉ ሁልጊዜ አሲዶችን በውሃ ውስጥ ማፍሰስ እንደሚገባ ይናገራሉ፤ ምክንያቱም ይህ ኃይለኛ ርጭትን ይከላከላል። አማራጭ A አስተማማኝውን ቅደም ተከተል ስለሚገለብጥ አሳሳች ነው።

11.2 Common Laboratory Equipment

- **Beaker:**
  - Common container for mixing, stirring, and heating chemicals.
  - Usually has spouts for pouring and volume markings, but these are not precise for measurement.
  - Comes in various sizes.
  - Can be covered with a watch glass to prevent contamination or splashing.

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- **Other Equipment (from Figure 2.7):**
  - Crucible tongs
  - Utility clamp
  - Clay triangle
  - Wire gauze
  - Test tube holder
  - Spatula
  - Test tube rack
  - Erlenmeyer flask
  - Florence flask
  - Graduated cylinder
  - Graduated pipet
  - Medicine dropper
  - Watch glass
  - Volumetric pipet
  - Funnel
  - Evaporating dish
  - Wash bottle
  - Test tube
  - Buret
  - Crucible & cover
  - Ring support
  - Bunsen burner
  - Ring stand
  - Scoopula
  - Test tube brush

📝 Practice Question 37 / የልምምድ ጥያቄ 37 (EUEE-style)

Question / ጥያቄ: Which statement about a beaker is correct? / ስለ ቢከር (beaker) የትኛው መግለጫ ትክክል ነው?

Options / አማራጮች:

- A. Its volume markings are not precise enough for accurate measurement [የመጠን ምልክቶቹ ለትክክለኛ መለኪያ በቂ ጥራት የላቸውም]
- B. It is the most precise device for measuring volume [መጠንን ለመለካት እጅግ ትክክለኛው መሣሪያ ነው]
- C. It must never be used for heating chemicals [ኬሚካሎችን ለማሞቅ ፈጽሞ ጥቅም ላይ መዋል የለበትም]
- D. It cannot be covered with a watch glass [በwatch glass ሊሸፈን አይችልም]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): A beaker is a common container for mixing, stirring and heating chemicals, and its volume markings are not precise for measurement. Option B is the main distractor, because precise volume work uses devices such as a burette, pipette, graduated cylinder or volumetric flask.

ማብራሪያ (አማርኛ): ቢከር ኬሚካሎችን ለመቀላቀል፣ ለማማሰልና ለማሞቅ የሚያገለግል የተለመደ መያዣ ሲሆን የመጠን ምልክቶቹ ለመለካት ትክክለኛ አይደሉም። አማራጭ B አሳሳች ነው፤ ትክክለኛ የመጠን ሥራ እንደ ቡሬት፣ ፒፔት፣ የተመረቀ ሲሊንደር ወይም የቮልዩሜትሪክ ብልቃጥ ያሉ መሣሪያዎችን ይጠቀማል።

12. Writing a Laboratory Report

A well-prepared laboratory report is the essential final step in performing an experiment, serving to communicate findings

effectively.

12.1 The Pre-laboratory Report

- **Purpose:** To ensure familiarity with the experimental procedure before the experiment begins, leading to more efficient use of laboratory time.
- **Completion:** Must be completed prior to starting the experiment.
- **Content:**
  - Answers to prelab questions (after reading the experiment's introduction and procedure).
  - Sometimes includes sample calculations to prepare students for data treatment.
  - Teachers may administer prelab quizzes instead.

12.2 The Laboratory Report

- **Format:**
  - A final draft, always written in ink or typed (if handwriting is illegible).
  - No erasures or crossed-out areas.
  - The initial draft should be in your laboratory notebook for two reasons:
    1. It prevents a messy final report, as initial attempts may require corrections.
    2. It serves as a backup, allowing easy rewriting if the final report is lost.
  - **Neatness:** Essential for making a good impression and indicating care for the subject matter.
- **Data Presentation:**
  - All data must include the correct significant figures and units.
  - Omission of units makes it hard to understand the magnitude of numbers.
  - Incorrect significant figures misrepresent the precision of data.
- **Answers to Questions:**
  - Should be complete, grammatically correct sentences.
  - Abbreviations should not be used in written answers.
- **Sample Computations:**
  - Must be labeled with their purpose (e.g., "mass of the liquid").
  - All numbers within computations must have correct units and significant figures.
- **Multi-page Reports:**
  - Should be stapled together or have the student's name and page number at the top right of each page (e.g., "Terhas Asgedom, page 2 of 4 pages").
  - Must be dated.

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📝 Practice Question 38 / የልምምድ ጥያቄ 38 (EUEE-style)

Question / ጥያቄ: Which statement about the final laboratory report is correct? / ስለ ላቦራቶሪ የመጨረሻ ሪፖርት የትኛው መግለጫ ትክክል ነው?

Options / አማራጮች:

- A. Erasures and crossed-out areas are acceptable in the final draft [በመጨረሻው ረቂቅ ውስጥ መሰረዝና የተሻገሩ ቦታዎች ተቀባይነት አላቸው]
- B. Units may be left out of the data when the numbers look obvious [ቁጥሮቹ ግልጽ ሲመስሉ ከመረጃው ውስጥ አሃዶች ሊቀሩ ይችላሉ]
- C. Abbreviations should be used in written answers to save time [ጊዜ ለመቆጠብ በጽሑፍ መልሶች ውስጥ ምህጻረ ቃላት መጠቀም አለባቸው]
- D. All data must include the correct significant figures and units [ሁሉም መረጃዎች ትክክለኛ ጠቃሚ አሃዞችና አሃዶች ሊኖራቸው ይገባል]

Correct Answer / ትክክለኛ መልስ: D

Explanation (English): In the report, all data must have the correct significant figures and units, because missing units hide the size of the numbers and wrong significant figures misrepresent precision. Option B is the main distractor, since the guide says omitting units makes the magnitude of numbers hard to understand.

ማብራሪያ (አማርኛ): በሪፖርቱ ውስጥ ሁሉም መረጃዎች ትክክለኛ ጠቃሚ አሃዞችና አሃዶች ሊኖራቸው ይገባል፤ ምክንያቱም የጎደሉ አሃዶች የቁጥሮችን መጠን ይደብቃሉ፣ የተሳሳቱ ጠቃሚ አሃዞች ደግሞ የመረጃውን ትክክለኛነት ያጣምማሉ። አማራጭ B አሳሳች ነው፤ መመሪያው አሃዶችን መተው የቁጥሮችን መጠን ለመረዳት እንደሚያስቸግር ይገልጻል።

12.3 Graphs

- **Purpose:** To present data visually (in picture form) for easier understanding and to show trends.
  - Can be used to follow a trend (drawing a best smooth curve, not just connecting dots).
  - Often used to show how well data fits a straight line (line can be visually estimated or mathematically computed).
- **Essential Features of a Good Graph:**
  1. **Axes:** Must be both numbered and labeled.
     - The horizontal axis (x-axis) is called the **abscissa**.
     - The quantity plotted on the vertical axis (y-axis) is mentioned first when speaking of graphing (e.g., “graph of y versus x”).
  2. **Title:** Every graph must have a clear title.
  3. **Data Points:** Never graphed as simple small dots. Use small circles, circles with dots, crosses, asterisks, or X’s to avoid points being lost or mistaken for ink blobs.
  4. **Lines:** Any lines drawn on the graph (e.g., best-fit lines) should be explained in the title or a legend.
  5. **Scales:** Axes scales should be adjusted to allow the graph to fill the page as much as possible.

📝 Practice Question 39 / የልምምድ ጥያቄ 39 (EUEE-style)

Question / ጥያቄ: Which of the following practices is **NOT** recommended when drawing a graph? / ግራፍ ሲሳል ከሚከተሉት ልማዶች ውስጥ የማይመከረው የትኛው ነው?

Options / አማራጮች:

- A. Numbering and labeling both axes [ሁለቱንም ዘንጎች መቁጠርና መሰየም]
- B. Plotting data points as simple small dots [የመረጃ ነጥቦችን እንደ ቀላል ትናንሽ ነጥቦች ማሳየት]
- C. Giving the graph a clear title [ለግራፉ ግልጽ ርዕስ መስጠት]
- D. Adjusting the axis scales so that the graph fills the page as much as possible [ግራፉ ገጹን በተቻለ መጠን እንዲሞላ የዘንግ ሚዛኖችን ማስተካከል]

Correct Answer / ትክክለኛ መልስ: B

Explanation (English): Data points should never be graphed as simple small dots, because they can be lost or mistaken for ink blobs; small circles, crosses, asterisks or X’s are used instead. Option D is the main distractor, since adjusting scales to fill the page is one of the recommended features of a good graph.

ማብራሪያ (አማርኛ): የመረጃ ነጥቦች እንደ ቀላል ትናንሽ ነጥቦች ፊጽሞ መሳል የለባቸውም፤ ምክንያቱም ሊጠፉ ወይም ከቀለም ጠብታ ጋር ሊምታቱ ይችላሉ፤ በምትኩ ትናንሽ ክቦች፣ መስቀሎች፣ ኮከብ ምልክቶች ወይም X ይጠቀማሉ። አማራጭ D አሳሳች ነው፤ ግራፉ ገጹን እንዲሞላ ሚዛኖችን ማስተካከል ከሚመከሩ የጥሩ ግራፍ ባህሪያት አንዱ ነው።

13. Measurements and Density: An Experiment Overview

This section sets up a practical experiment to familiarize students with scientific measurements, instrument use, and the evaluation of data quality.

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- **Core Skills:** Learning how to carry out scientific measurements properly using common instruments, and appreciating the significance of recorded data in terms of significant figures, precision, and accuracy.
- **Experiment Objectives:**
  - Practice using a balance for mass measurements.
  - Learn to measure volume using a graduated cylinder and a pipet.
  - Calibrate a pipet.
  - Determine the density of a metal bar and a salt solution using different methods.
  - Evaluate measurement results for precision (relative average deviation) and accuracy (relative error).
- **Density:**
  - One of an object's most fundamental and useful characteristics.
  - An **intensive property**: independent of the quantity of material measured, as it's the ratio of mass to volume.
- **Prerequisites:** Students should carefully study the Introduction to the laboratory manual, particularly sections on precision, accuracy, significant figures, and laboratory notebooks.

13.1 Materials and Chemicals

- Cylindrical metal bars (Al, Cu, brass), approximately 5-1 cm diameter.
- Measuring rules (graduated in mm).
- 20 or 25 mL transfer pipets.
- 50 mL beaker.
- Graduated cylinders (10 and 50 mL or 100 mL).
- 125 mL Erlenmeyer flask.
- Stopper.
- Thermometers.
- Balances with precision to 1 mg.
- Saturated salt solutions (NaCl and/or KCl) - approx. 36 g NaCl / 100 mL.
- Estimated Time: 2-3 hours.

13.2 Safety Precautions

- Review general safety rules.
- **Special care:** When inserting the metal bar into a graduated cylinder, do *not* drop it to avoid breaking the glass cylinder.
- **Pipeting:** Always use a suction device; never suction by mouth.

13.3 Procedures

- **General Instructions:**
  - Record all measurements in your laboratory notebook **in ink**.
  - Proper use of a sensitive balance and pipeting technique are critical. Instructor demonstration is important.
  - If an instruction says "Repeat… twice," it means performing that portion a total of three times (initially + two repetitions).

Part I: Mass Measurements (Section A)

1. Zero the balance after cleaning the pan.
2. Measure the mass of a clean, dry 50 mL beaker to the nearest 0.001 g.
3. Record the observation directly into the lab notebook in ink.
4. Remove the beaker, clean the pan, and zero the balance again.
5. Weigh the same beaker (from step 2) and record the result.
6. Repeat steps 4 and 5 one more time (total of three weighings).
7. Calculate the average mass of the beaker from the three measurements.

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8. Repeat steps 4 and 5 using a second balance (just one weighing).
9. Repeat steps 4 and 5 using a third balance (just one weighing).

Part II: Volume Measurements (Section B)

- **Use of a Pipet:**
  - Most volumetric pipets are "to deliver" (TD), meaning a small amount of liquid remains in the tip after transfer.
  - **Safety:** Never pipet by mouth; always use a suction device.
  - **Technique:**
    - Use a clean (but not necessarily dry) 20 or 25 mL pipet.
    - Rinse the pipet several times with small portions of the liquid to be transferred.
    - Squeeze air from suction device, create a tight seal on the pipet.
    - Dip pipet tip below liquid surface (avoid touching bottom, as a chipped tip causes error).
    - Release suction to fill pipet 1-2 cm above the calibration line.
    - Quickly remove suction device and cover the end with your index finger.
    - Wipe the outside of the tip.
    - Drain liquid until the meniscus rests exactly on the calibration line (touch tip to wall of source container).
    - Hold pipet over sample container and allow it to drain (avoid splashing).
    - When nearly empty, touch tip to container wall for an additional 20 seconds.
    - Do not blow out remaining liquid.
- **Steps for Water Volume Measurement:**
  1. Measure laboratory temperature (density of water at this temperature will be provided).
  2. Use the 50 mL beaker from Section A. Its average dry mass can be used.
  3. Measure 20 or 25 mL of water (depending on pipet size) into the beaker using the pipet.
  4. Record the volume of water from the pipet with appropriate significant figures.
  5. Record the number of significant figures in the volume measurement.
  6. Weigh the beaker and water to the nearest mg (± 0.001 g).
  7. Calculate the mass of water in the beaker.
  8. Use the mass and density of water to determine the actual volume of water measured.
  9. Repeat steps 3-8 using a 50 or 100 mL graduated cylinder instead of the pipet.
  10. Repeat steps 3-8 again using a graduated 50 mL beaker to measure the water.

Part III: Density of a Metal Bar (Section A)

Use the same metal bar for all trials.

1. **Mass Measurement:**
   - Zero the balance.
   - Weigh the metal bar on a balance sensitive to 0.001 g.
   - Repeat the entire weighing operation twice (total of three weighings), ensuring not to be influenced by prior measurements. Zero the balance before each measurement.
2. **Volume Determination (Three measurements for each method):**
   - **Method I (Water Displacement):**
     - Insert the bar into a graduated cylinder filled with enough water to immerse the bar (do not drop!).
     - Record the initial and final water levels as precisely as possible, reading the lowest point of the meniscus and estimating one digit beyond the smallest scale division.
     - Discard water and repeat twice with different initial water volumes.
     - Calculate the average density of the bar.
   - **Method II (Dimensions):**
     - Measure the dimensions (height and diameter) of the cylindrical bar with a measuring stick ruled in

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centimeters.
    - Repeat these measurements twice.
    - Calculate the volume of the bar using the formula for a cylinder: $V = \pi r^2 h$
    - Calculate the average density of the bar.
3. **Error Analysis:** For each method, determine the **relative error** by comparing your result with an accepted value (provided by instructor or reference).

Part IV: Density of a Salt Solution (Section B)

1. **Method 1 (Volumetric Pipet):**
    - Weigh a 125 mL Erlenmeyer flask and stopper.
    - Using a clean 20.00 or 25.00 mL volumetric pipet, transfer the salt solution into the flask and reweigh.
    - Repeat this measurement twice with different samples of the same solution.
    - Calculate the average density of the salt solution.
2. **Method 2 (Graduated Cylinder):**
    - Weigh an empty, dry 10 mL graduated cylinder.
    - Fill with about 9-10 mL of salt solution, record the volume precisely, and reweigh.
    - Repeat this measurement twice with different samples.
    - Calculate the average density of the salt solution.
3. **Precision Analysis:** For each method, determine the **relative average deviation** of your results. Compare methods to determine which is more precise.

13.4 Disposal

- **Salt solutions:** Follow teacher's instructions:
    - **Recycle:** Return to original container.
    - **Treatment/Disposal:** Dilute 1:10 with tap water and flush down sink.
    - **Disposal:** Put in a waste bottle labeled "inorganic waste."

13.5 Questions

1. From your data, calculate the volume occupied by 100 g of the:
    - Salt solution
    - Metal bar
2. From your answers to question 1, determine whether the metal bar or the salt solution occupies the larger volume. Explain your answer in the context of the densities of solids and liquids in general.
3. Define the terms **precision** and **accuracy** in such a way as to distinguish between them.
4. Are your results for the metal bar more precise or more accurate? Explain.
5. From your data for the salt solution, evaluate the two methods in terms of their precision. Which method should lead to greater precision? Which method actually is more precise? Explain.

13.6 Prelab Measurements and Density

- **Given Data (Cylindrical Metal Bar):**

Trial	1	2	3
Height (cm)	6.50	6.45	6.44
Diameter (cm)	1.25	1.26	1.22
Mass (g)	46.683	46.332	47.014

- **Prelab Calculations (Show formula, substituted numbers, and result):**
    1. Calculate the average density of the bar.

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2. Calculate the percent relative average deviation of the measurements.
3. If the accepted value for the density of the bar is 6.70 g/cm^3, what is the percent relative error?
4. Are these measurements more precise or more accurate? Explain.
5. What is the purpose of repetition in measurements?

📝 Practice Question 40 / የልምምድ ጥያቄ 40 (EUEE-style)

Question / ጥያቄ: Using the Trial 1 data of the prelab table (height 6.50 cm, diameter 1.25 cm, mass 46.683 g) and V = πr^{2}h, what is the density of the cylindrical metal bar? / በቅድመ ላቦራቶሪ ሰንጠረዥ የሙከራ 1 መረጃ (ቁመት 6.50 cm፣ ዲያሜትር 1.25 cm፣ ክብደት 46.683 g) እና V = πr^{2}h በመጠቀም የሲሊንደሩ ብረት ጥግግት ስንት ነው?

Options / አማራጮች:

- A. 5.85 g/cm^3 [5.85 g/cm^3]
- B. 1.46 g/cm^3 [1.46 g/cm^3]
- C. 3.66 g/cm^3 [3.66 g/cm^3]
- D. 7.98 g/cm^3 [7.98 g/cm^3]

Correct Answer / ትክክለኛ መልስ: A

Explanation (English): The radius is half of the diameter, r = 1.25 / 2 = 0.625 cm, so V = π × (0.625)^2 × 6.50 ≈ 7.98 cm^3 and d = m / V = 46.683 g / 7.98 cm^3 ≈ 5.85 g/cm^3, given to three significant figures. Option B is the main distractor: it uses the diameter (1.25 cm) as the radius, which makes the volume about 31.9 cm^3 and the density about 1.46 g/cm^3; option D is just the volume, not the density.

ማብራሪያ (አማርኛ): ራዲየስ የዲያሜትር ግማሽ ነው፣ r = 1.25 / 2 = 0.625 cm፤ ስለዚህ V = π × (0.625)^2 × 6.50 ≈ 7.98 cm^3 ሲሆን d = m / V = 46.683 g / 7.98 cm^3 ≈ 5.85 g/cm^3 ነው፣ በሦስት ጠቃሚ አሃዞች። አማራጭ B አሳሳች ነው፤ ዲያሜትሩን (1.25 cm) እንደ ራዲየስ ይጠቀማል፤ ይህ መጠኑን ወደ 31.9 cm^3 ያህል፣ ጥግግቱን ደግሞ ወደ 1.46 g/cm^3 ያህል ያደርገዋል፤ አማራጭ D መጠኑ እንጂ ጥግግቱ አይደለም።

14. Key Terms

- Accuracy
- Calibration
- Derived units
- Decimal places
- Error
- Exact numbers
- Laboratory safety
- Laboratory equipment
- Measurement
- Measured numbers
- Significant figures
- Scientific notations
- Scientific method
- SI units
- Prefixes
- Precision
- Uncertainty

15. Unit Summary

- The International System (SI) uses a specific selection of metric units, comprising seven base units combined with prefixes for various sizes.
- Derived units are formed by substituting base units into defining equations for quantities like volume or density. SI units are universally used in science.
- Scientific notation facilitates handling very large and very small quantities.
- Most measured quantities are inherently inexact. The number of significant figures communicates the exactness (or uncertainty) of a measurement.
- Accuracy describes how closely a measured value matches the true value.
- Precision describes how closely repeated measurements of the same quantity agree with each other.
- Chemistry is an experimental science, with its facts derived from experiments.
- The scientific method is a systematic approach to research, starting with observation and measurements, leading to the development and testing of hypotheses, laws, and theories.
- Repetition is crucial in experimental measurements for ensuring confidence in the significance of the results.

Answer Key

Question #	Correct letter	Subtopic
1	C	1. Introduction to Measurement
2	B	1. Introduction to Measurement
3	D	1. Introduction to Measurement
4	A	2. SI Units
5	D	2. SI Units
6	B	2.1 Heat and Temperature
7	C	2.1 Heat and Temperature
8	A	2.1 Heat and Temperature
9	C	2.1 Heat and Temperature
10	C	3. Derived Units
11	B	3.1 Volume
12	A	3.2 Density
13	D	3.2 Density
14	C	3.2 Density
15	D	4. Common Prefixes Used in SI Units
16	C	4. Common Prefixes Used in SI Units
17	A	5.1 Categories of Uncertainty
18	B	5.1 Categories of Uncertainty
19	C	5.2 Calculating Uncertainties
20	D	5.2 Calculating Uncertainties
21	B	5.2 Calculating Uncertainties
22	A	6. Precision and Accuracy
23	D	6. Precision and Accuracy
24	B	6. Precision and Accuracy
25	C	7.1 Rules for Determining Significant Figures
26	A	7.1 Rules for Determining Significant Figures
27	D	7.1 Rules for Determining Significant Figures
28	B	7.2 Significant Figures in Calculations

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29	C	7.2 Significant Figures in Calculations
30	A	7.2 Significant Figures in Calculations
31	C	8. Scientific Notation and Decimal Places
32	B	8. Scientific Notation and Decimal Places
33	D	9. Chemistry as Experimental Science
34	A	10. The Scientific Method
35	C	10. The Scientific Method
36	B	11.1 Laboratory Safety Rules
37	A	11.2 Common Laboratory Equipment
38	D	12.1–12.2 Writing a Laboratory Report
39	B	12.3 Graphs
40	A	13. Measurements and Density Experiment

Placement Table

Subtopic	Question numbers	Count
1. Introduction to Measurement	1–3	3
2. SI Units	4–5	2
2.1 Heat and Temperature	6–9	4
3. Derived Units	10	1
3.1 Volume	11	1
3.2 Density	12–14	3
4. Common Prefixes Used in SI Units	15–16	2
5.1 Categories of Uncertainty	17–18	2
5.2 Calculating Uncertainties	19–21	3
6. Precision and Accuracy	22–24	3
7.1 Rules for Determining Significant Figures	25–27	3
7.2 Significant Figures in Calculations	28–30	3
8. Scientific Notation and Decimal Places	31–32	2
9. Chemistry as Experimental Science	33	1
10. The Scientific Method	34–35	2
11.1 Laboratory Safety Rules	36	1
11.2 Common Laboratory Equipment	37	1
12.1–12.2 Writing a Laboratory Report	38	1
12.3 Graphs	39	1
13. Measurements and Density Experiment	40	1
Total	**1–40**	**40**

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